

Lab Assignment 6: Spam Email Detection

CSAI2017P: Applied Machine Learning (Lab)

Experiment 8 (Unit VI) • CO2, CO4

Instructor: Dr. Tofik Ali**Due:** announced in the lab**Student Name:** _____**SAP ID:** _____**Batch:** _____**Lab quiz:** LQ6, after submission

Objective

- Turn free text into numeric features with bag-of-words and TF-IDF.
- Fit and interpret a Naive Bayes text classifier.
- Choose a metric appropriate to a task where one error type is far worse.

Datasets used in this assignment

C2 (SMS Spam Collection).

Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA06_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			10

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 ($2 \times 2 = 4$)**A1. Text to vectors [2 marks]**

Dataset: C2.

- (a) Load the corpus and report the number of messages and the spam rate.
- (b) Fit a `CountVectorizer` and report the vocabulary size.
- (c) Show the ten most frequent tokens and state in two lines why they are useless for classification.

A2. Naive Bayes [2 marks]

Dataset: C2.

- (a) Fit `MultinomialNB` on the counts and report accuracy, precision, recall and F1 on the spam class.
- (b) Print the confusion matrix.
- (c) State in two lines which words the model finds most spam-indicative.

A3. Counts versus TF-IDF [2 marks]

Dataset: C2.

- (a) Refit the same classifier on `TfidfVectorizer` features.
- (b) Report the same four metrics and compare with A2 in a small table.
- (c) Explain in two lines what the IDF term is doing.

Section B (Medium) — attempt any 1 of 2 ($1 \times 4 = 4$)**B1. The cost of a false positive [4 marks]**

Dataset: C2.

- (a) For your best model, report precision and recall on the spam class at thresholds 0.3, 0.5, 0.7 and 0.9.
- (b) Plot the precision-recall curve and mark the four points.
- (c) A real filter sends spam to a junk folder. Argue in 6–8 lines which error is worse and pick an operating threshold.

B2. n-grams and vocabulary size [4 marks]

Dataset: C2.

- (a) Fit the same classifier with `ngram_range` of (1,1), (1,2) and (1,3).
- (b) Report vocabulary size, F1 and fit time for each in one table.
- (c) Explain in 4–6 lines what bigrams add, and at what point the extra features stop paying for themselves.

Section C (Hard) — compulsory (2)

C1. Break your own filter

[2 marks]

Dataset: C2.

- (a) Write three spam messages by hand that your model classifies as ham.
- (b) In 4–6 lines, explain what property of bag-of-words you exploited and one defence against it.

End of Lab Assignment 6

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.